HTML 5

1.1 What and Why HTML 5?

This document is prepared to understand html5 and its tags to prepare more attractive UI with extra ordinary features.

HTML5 is backwards compatible. It contains all the features of the HTML4 spec, although with a few changes and improvements. But it also contains much more extra stuff for building dynamic web applications and creating higher quality markup, such as:

- New semantic elements to allow us to define more parts of our markup unambiguously and semantically rather than using lots of classes and Ids.
- New elements and APIs for adding video, audio, scriptable graphics, and other rich application type content to our sites.
- New features for standardizing functionality that we already built in needs of an individual customer.

HTML5 has been deliberately built to compete with proprietary plug-ins such as Flash and Silverlight. It looks like such technologies are increasingly taking a back seat. And as you will see, a lot of HTML5 has good support across all modern browsers and therefore is usable right now in your projects. It can even be made to work in older browsers with a bit of JavaScript modification.

1.2 It's All in the head(not yours!It's HTML head tag)!!(Doctype Introduction)

Before we start with on creating some good “Hello World” web pages, let's take a look at some changes we will encounter,specifically changes we'll be making in the <head> as we move toward HTML5.

In this section, we'll introduce the new HTML5 DOCTYPE, explain how to declare languages in HTML5, and introduce a new, simpler meta charset attribute.

In html 4 the doctype and character coding(charset) which you use is

```
<!DOCTYPE html PUBLIC "-//W3C//DTD HTML 4.01 Transitional//EN"
"http://www.w3.org/TR/html4/loose.dtd">
<html>
<head>
<meta http-equiv="Content-Type" content="text/html; charset=ISO-8859-1">
<title>HTML 4 doctype and charset</title>
</head>
```

```
<body>

</body>
</html>
```

Now for HTML 5 it s much more simpler!!

```
<!DOCTYPE html>
<html lang="en">
   <head>
      <meta charset="UTF-8" />
      <title>HTML 5 doctype and charset</title>
   </head>
<body>

</body>
</html>
```

You don't really need to know what character encoding is in detail, but basically, it defines the set of valid human language characters that can be used in the document. It is safest to stick to UTF-8, a universal character set that allows all characters from all languages, just about.

1.3 Hello World Using HTML 5

“Hello world” in html 4 is mentioned below and then “Hello World” in html 5 is mentioned.

HTML 4 “Hello World”

```
<!DOCTYPE HTML PUBLIC "-//W3C//DTD HTML 4.01//EN"
      "http://www.w3.org/TR/html4/loose.dtd">
<html>
<head>
   <meta http-equiv=content-type content="text/html; charset=UTF-8">
   <title>Hello World! HTML 4 loose</title>
</head>
<body>
   <p>Hello World!</p>
</body>
</html>
```

HTML 5 “Hello World”

```
<!DOCTYPE html>
   <meta charset=UTF-8>
   <title>Hello World!</title>
   <p>Hello World!
```

If we use strict style then it is

```
<!DOCTYPE html>
<html lang="en">
   <head>
      <meta charset="UTF-8" />
      <title>Hello World!</title>
   </head>
   <body>
      <p>Hello World!</p>
   </body>
</html>
```

1.4 HTML 5 Tags

1.4.1 Structural Element

Here we will first look few of the structural elements which will help you to define basci structure of your page.

- **Article**

Tag name : Article(<article>)

Description:

The HTML5 Article element (<article>) represents independent (a web document, page, site or application) content, for example a forum post, a magazine or newspaper article, a blog entry, a user-submitted comment, an interactive widget or gadget.

The HTML5 Article Element contains either actual content or contains several different section elements or one or more article elements within it.

If it contains article element within it, the content of the inner articles supposed to be related to the outer article's content.

Using **pubdate** attribute of time element, date and time of publication of an article can be described.

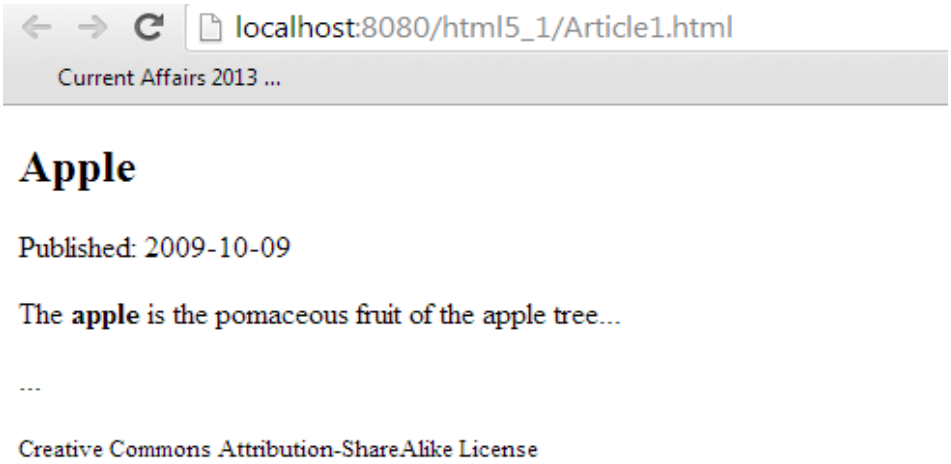
Article element can contain Text or embedded content.

Attributes : Supports all global attributes and event attributes defined in conventional html.

Code :

```
<!DOCTYPE html>
<html>
<head>
<meta http-equiv="Content-Type" content="text/html; charset=ISO-8859-1">
<title>HTML5 article example</title>
</head>
<body >
<article>
  <header>
    <h1>Apple</h1>
    <p>Published: 2009-10-09</time></p>
  </header>
  <p>The <b>apple</b> is the pomaceous fruit of the apple tree...</p>
  ...
  <footer>
    <p><small>Creative Commons Attribution-ShareAlike License</small></p>
  </footer>
</article>
</body>
</html>
```

Output:



Browser Support:

Element					
<article>	6.0	9.0	4.0	5.0	11.1

- ## Section

Tag Name: <Section>

Description:
The *HTML Section Element* (<section>) represents a generic section of a document, i.e., a thematic grouping of content, typically with a heading. Each <section> should be identified, typically by including a heading (h1-h6 element) as a child of the <section> element.

Attributes : Supports all global attributes and event attributes defined in conventional html.

Code:

```
<article>
<section>
        <h4>What We Do</h4>
        <p>We protect sharks...</p>
```

```
        </section>
        <section>
                <h4>News Items</h4>
                <p>Man eats shark...</p>
        </section>
        <section>
        <h4>Contact Us</h4>
                <p>At the beach...</p>
                </section>
        </article>
```

Output:

What We Do
We protect sharks...

News Items
Man eats shark...

Contact Us
At the beach...

Real Time Application:
If you want to make a one page website in html5 and also you want to provide navigation across that one page like scrolling effect than it is useful.

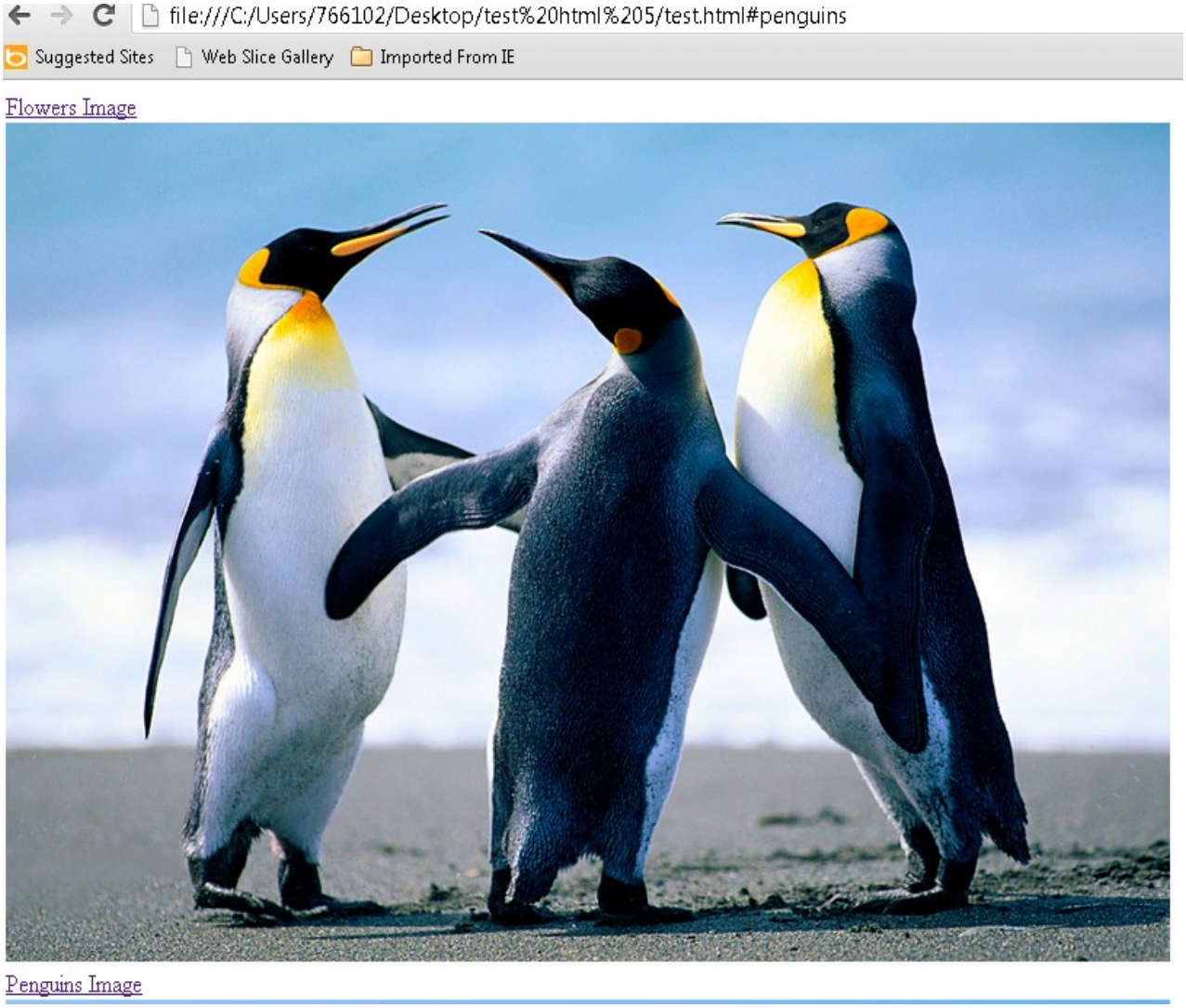
Code:

```
        <!DOCTYPE HTML>
<!DOCTYPE HTML>
<html>
<head>
<meta http-equiv="Content-Type" content="text/html" charset="ISO-8859-1">
<title>Section Tag Usage</title>
</head>

<body>
<a href="#flowers">Flowers Image </a>

<section id="penguins">  <img src="C:\Users\Public\Pictures\Sample Pictures\Penguins.jpg"
height="550px" width="800px"/></section>
```

```
<a href="#penguins">Penguins Image</a>
<section id="flowers" ><img src="C:\Users\Public\Pictures\Sample Pictures\Tulips.jpg"
height="550px" width="800px"/></section>

</body>
</html>
```

Output:

For Section id=”penguins”
When you click on Flower Image Link, page will be navigated to section id=”flowers” and when you click on Penguins Image link, page will be navigated to section id =”penguins”.



For Section id=”flowers”

Penguins Image

Browser Support :

Element					
<embed>	Yes	Yes	Yes	Yes	Yes

- ## Header

Tag Name : <header>

Description :

- "header" is the new tag included in HTML5. It is used to specify header information such as Title, Introductory Content for whole document or for particular section.

- This typically includes the heading (an <h1>–<h6> element or an <hgroup> element), but can also contain other content, such as a <nav> element or navigational links, a table of contents, a search form, or any relevant logos. It can't contain <footer> or another <header>

- <header>'s defining purpose is to contain introductory or navigational aids, rather than define their placement (headers are usually at the top of a section, but not necessarily).

- <footer> is for related links, copyright, and author information. A <footer> is usually at the end of a section but can appear at the top and more than once (for example, a "Top of page" link). For some content, there is some crossover in which of these elements to use—author information would be fine in either.

- <header> and <footer> seem to suggest the page header and footer familiar to designers. Don't let first impressions fool you, however, as these elements apply to the current sectioning element. If that element is <body>, then the <header> or <footer> does apply to the whole page. But if it's another sectioning element, then <header> or <footer> only applies to that element. Think Russian nested dolls…in cute little angle brackets.

- In earlier versions, we used and css to specify any header informations.

- Now, with HTML5 , it is enough to use only this tag.

- A document may contain any number of header tags.

- It should not be nested inside footer, another header tag or address tags.

Attributes : Supports all global attributes and event attributes defined in conventional html.

Code :

```
<!DOCTYPE html>
```

```
<html>
<body>
        <article>
                <header>
                        <h1>Header of outer Article</h1>
                                <p>HTML5 makes life easy..</p>
                </header>
                <article>
                        <header>
                                <h1>Header of Inner Article </h1>
                                        <p>HTML5 make life easy..</p>
                        </header>
                </article>
        </article>
</body>
</html>
```

Output:

Real time Application: To specify heading information of any document in your web page.

Browser Support :

Element					
<header>	6.0	9.0	4.0	5.0	11.1

- ## Footer

Tag Name : <footer>

Description :

• "footer" is the new tag included in HTML5. It is used to specify footer information such as Author Name, Contact Address, Copyright Information for whole document or for a particular section.

• In earlier versions, we used and css to specify any footer informations.

• Now, with HTML5 , it is enough to use only this tag.

• A document may contain any number of footer tags.

• For specifying contact information, we can use combination of footer and address tag.

Attributes: Supports all global attributes and event attributes defined in conventional html.

Code :

```
<!DOCTYPE html>
<html>
<body>
        <footer>

                <h4>Footer Example:</h4>

                <address>
                        You can contact us at:<br>
                        Spec Enterprises<br>
                        123, Whitefields Road,<br>
                        Bengaluru, Karnataka
                 </address>
        </footer>

</body>
</html>
```

Output:

Footer Example:

You can contact us at:
Spec Enterprises
123, Whitefields Road,
Bengaluru, Karnataka

Real time Application: To specify address information of any document in your web page.

Browser Support :

Element					
<footer>	6.0	9.0	4.0	5.0	11.1

- ## Nav

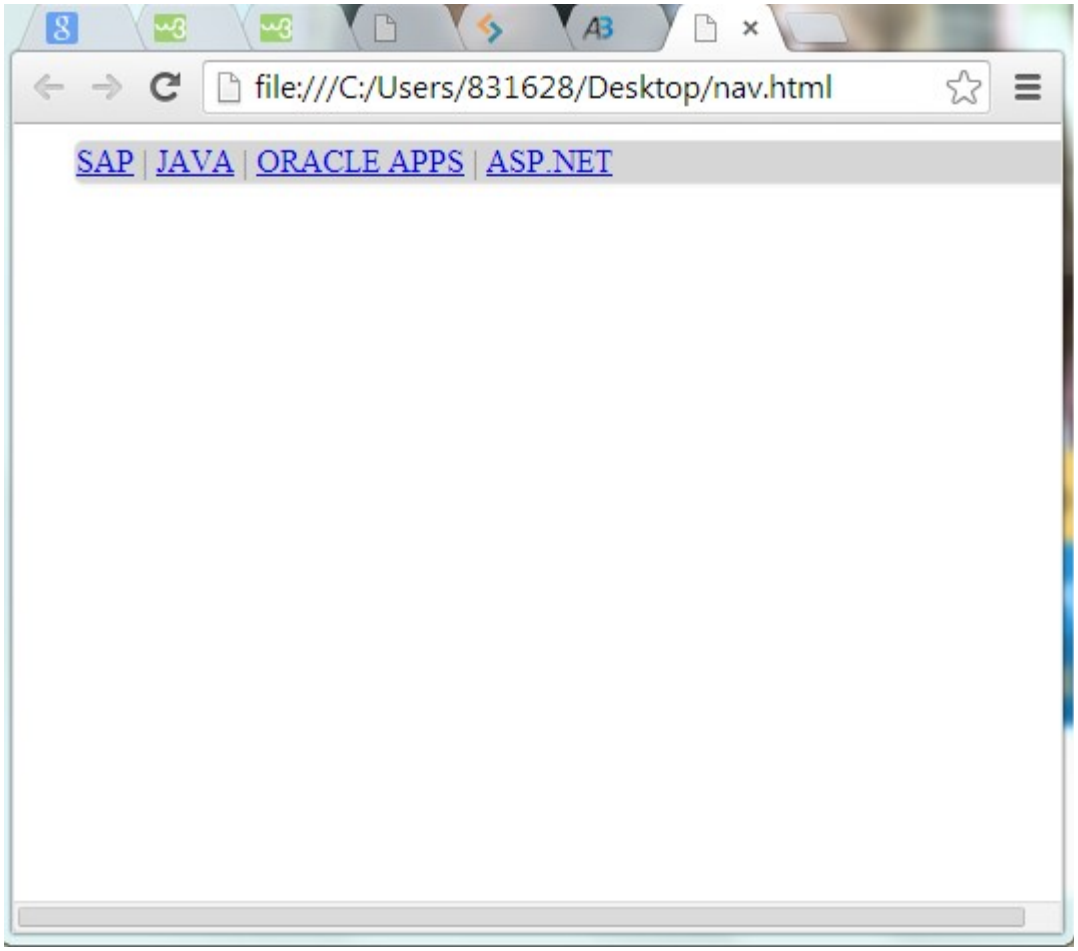
Tag Name: <nav>

Description :
The <nav> tag defines the set of the navigation links. This <nav> tag is basically intended for providing the support for navigation.
Attributes : Supports all global attributes and event attributes defined in conventional html.

Code: To create a navigation bar containing links for various pages.

```
<nav>
        <a href="/html/">SAP</a> |
        <a href="/css/">JAVA</a> |
        <a href="/js/">ORACLE APPS</a> |
        <a href="/jquery/">ASP.NET</a>
</nav>
```

Output:



Real time Application : In most of the sites navigation bar is provided to navigate through the available pages on the site.

Browser Support:

Element					
<nav>	6.0	9.0	4.0	5.0	11.1

- ## Aside

TagName: <aside>
Description:

The <aside> tag defines some content aside from the content it is placed in.Information either unrelated or loosely related to the main content of a page should go in aside tags.

Attributes : Supports all global attributes and event attributes defined in conventional html.

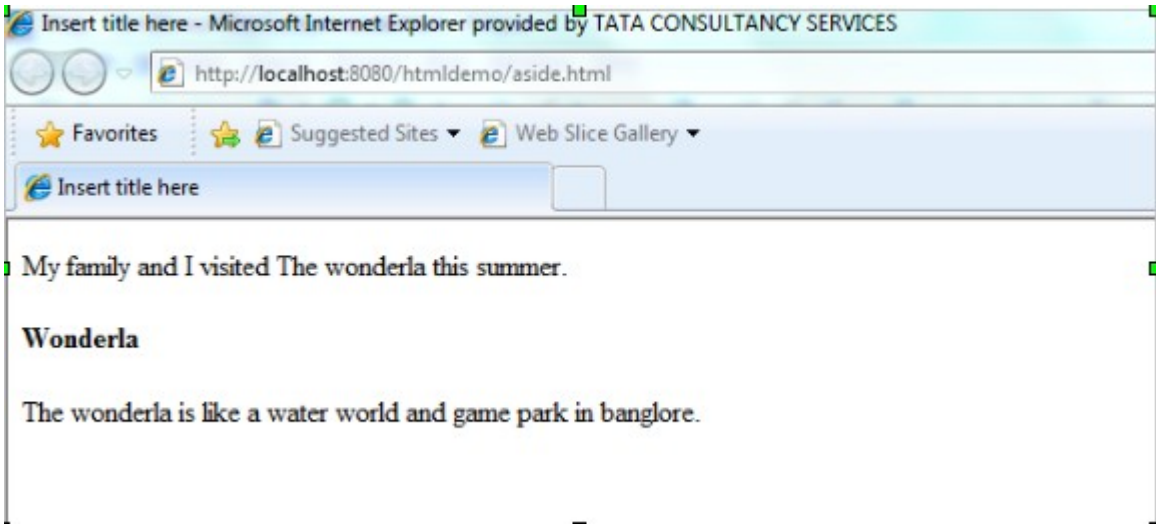
Real Time Application:

This element can be used for some effects like pull quotes or sidebars, for *advertising*, for groups of nav elements, and for other content that is considered separate from the main content of the page.

Code:

```
<body>

    <p>My family and I visited The wonderla this summer.</p>

    <aside>

    <h4>wonderla</h4>

    <p>The wonderla is like a water world and game park in banglore.</p>
    </aside>

</body>
```

Output:



Browser Support:

Element					
<aside>	6.0	9.0	4.0	5.0	11.1

- **Figure**

Tag name : <figure>

Description :
The <figure> tag is used to contain specific contents like illustrations, diagrams, photos, code listings, etc.

While the content of the <figure> element is related to the main flow, its position is independent of the main flow, and if removed it should not affect the flow of the document.

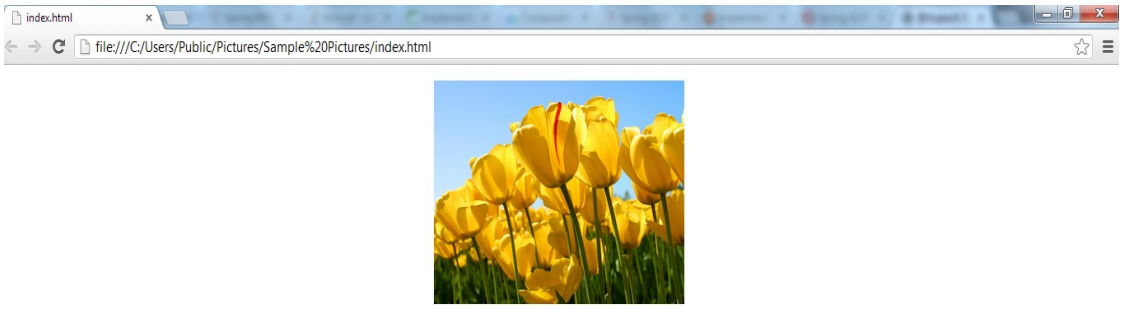
Attributes :

Global Attributes	All are supported
Event Attributes	Media,Mouse,Keyboard,Window and Form events.e.g. onblur, onclick, ondragleave, onkeydown, onerror, onplay, onreset, onfocus,onload, onundo, etc.

Code : To include an image in center of page

```
<!DOCTYPE html>
<html>
<body>
<figure align="center">
  <img src="tulip.jpg" alt="The Amazing Flower" width="300" height="250">
</figure>
</body>
</html>
```

Output :



Browser Support :

The numbers in the table specify the first browser version that fully supports the <figure> tag.

Browser	Chrome	IE	Firefox	Safari	Opera
Version	6	9	4	5	11.1

- ## Fig Caption

Tag name : <figcaption>

Description : The <figcaption> tag defines a caption for a <figure> element.
The <figcaption> element can be placed as the first or last child of the <figure> element.

Attributes :

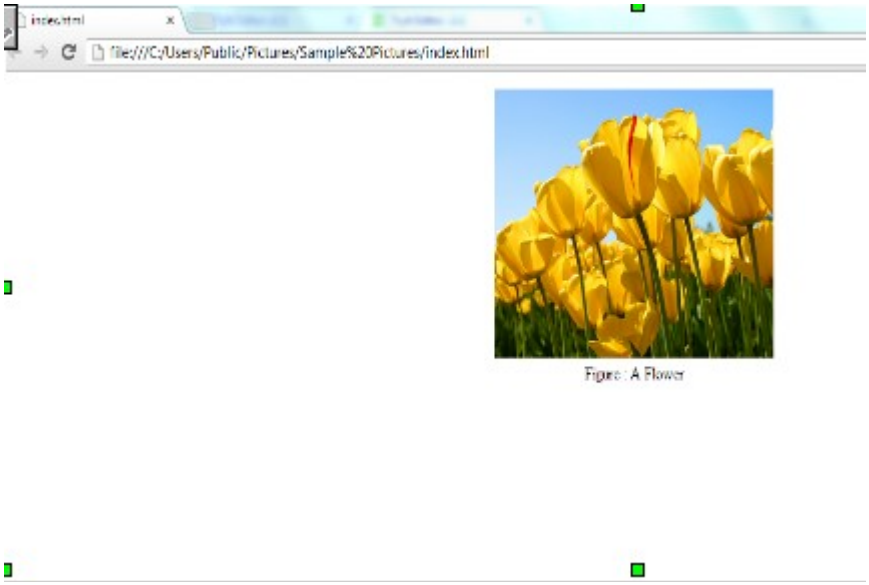
Global Attributes	All are supported
Event Attributes	Media,Mouse,Keyboard,Window and Form events.e.g. onblur, onclick, ondragleave, onkeydown, onerror, onplay, onreset,

	onfocus,onload, onundo, etc.

Code : To add a caption to some figure.

```
<!DOCTYPE html>
<html>
<body>
<figure align="center">
  <img src="tulip.jpg" alt="The Amazing Flower" width="300" height="250">
  <figcaption>Figure : A Flower</figcaption>
</figure>
</body>
</html>
```

Output :



Browser Support :

Browser	Chrome	IE	Firefox	Safari	Opera
Version	6	9	4	5	11.1

1.4.2 Ex-Presentational Element Got Semantic Meaning

First of all what is semantic meaning in terms of HTML?What is semantic HTML?Well the answer is very simple. Semantic HTML or semantic markup is HTML that give some meaning

to the web page rather then just presentation.

For example, a <p> tag indicates that the enclosed text is a paragraph. This is both semantic and presentational, because people know what a “paragraph” is and browsers know how to display them.

However, tags like <b> and <i> are not semantic, because they define only how the text should look (bold or italic) and do not provide any meaning to the text.

- **Giving the <i> and <b> Elements New Semantic Meanings**

In HTML 4, <i> and <b> were font style elements . However, they now have semantic meaning

The <i> Element

The <i> element represents a span of text in an alternate voice or mood, or otherwise offset from the normal prose in a manner indicating a different quality of text.

Things that are typically italicized include foreign words (using the attribute lang=""), taxonomic and technical terms, ship names, inline stage directions in a script, some musical notation, and when representing thoughts or hand-written text inline.

It is a good idea to use the class attribute to identify why the element is being used, so that if the presentation needs to change at a later date, it can be done selectively with style sheets. However it will not make any kind of difference in text presentation.

You can use <i class="voiceover"> to indicate a voiceover (alternate mood).Output will be the same as html 4 , and the reason to indicate class attribute is just explained in the above paragraph.

Similarly, <i lang="ja-latn"> to indicate transliterated prose from a foreign language (with lang="ja-latn" being transliterated Japanese).

And Using <i class="taxonomy"> for taxonomic names.

Again to clarify output of <i> tag will remain same as it is there in html4 no matter which class you are using(voiceover,taxanomy or lang attribute).

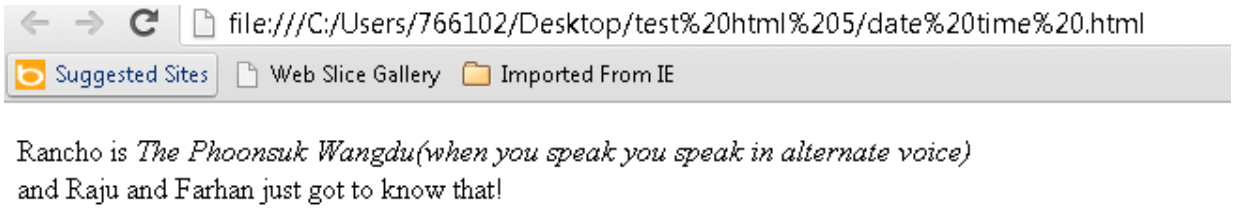
Attributes : Supports all global attributes defined in conventional html.

Code :

```
<!DOCTYPE html>
<html>
<body>

        <p>Rancho is  <i class="voiceover">The Phoonsuk Wangdu(when you speak you
```

```
speak in alternate voice)</i> and Raju and Farhan just got to know that!</p>
</body>
</html>
```

Output :



The <b> Element

The <b> element represents a span of text to which attention is being drawn for utilitarian purposes without conveying any extra importance and with no implication of an alternate voice or mood.

For <b> that should merely look different, there is no requirement to use font-style: bold;. Other styling could include a round-cornered background, larger font size, different color, or formatting such as small caps.

<b class="character"> is used to indicate who's speaking or narrating in any document.

Again it will not make any difference in terms of presentation, it looks same as it looks in html 4.Reason to write class name is same as it is for <i> tag.

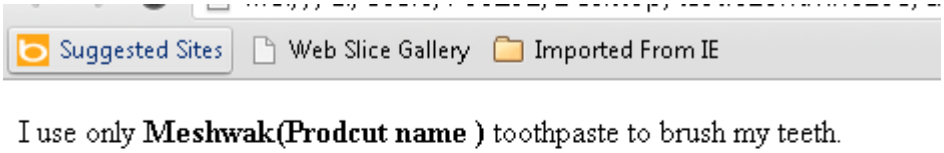
You could use <b> for product names in a review, key words in a document abstract, and the initial text on a complex or traditionally designed page.

Only use <b> when there are no other more suitable elements, such as for text with semantic importance; for emphasized text (text with "stress emphasis"); <h1>-<h6> for titles; and <mark> for highlighted or marked text. Again, remember to use the class attribute to identify why the element is being used, making it easy to restyle a particular use

Code :

```
<!DOCTYPE html>
<html>
<body>

        <p>I use only <b>Meshwak(Prodcut name )</b> toothpaste to brush my teeth.</p>
</body>
</html>
```

Output : .



- **The and Elements**

While and remain pretty much the same, there has been a slight realignment in their meanings. In HTML 4, they meant emphasis and strong emphasis. Now their meanings have been differentiated into representing stress emphasis (such as something you'd pronounce differently) and representing importance.

The Element

The em element represents stress emphasis of its contents.

The "stress" being referred to is linguistic; if spoken it would be emphasized pronunciation on a word that can change the nuance of a sentence. For example "Call a doctor now!" stresses doctor, perhaps in reply to someone asking "She looks thirsty, should I call a homeopath?" In comparison, "Call a doctor now!" changes the emphasis to doing so quickly.

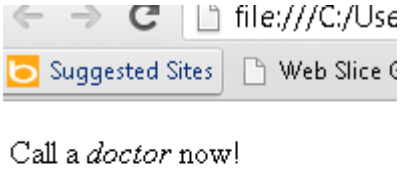
Again please do not confuse , representation of text is similar to HTML4 but it has just added semantic to it in HTML5.

Use instead to indicate importance and <i> when you want italics without implying emphasis. If an element occurs in text that is already emphasized with , the level of nesting represents the relative level of emphasis.

Code :

```
<!DOCTYPE html>
<html>
<body>

        <p>Call a <em>doctor</em> now!</p>
</body>
</html>
```

Output :



The Element

The strong element represents strong importance for its contents.

There's not much more to say, really—it's the we all know so well. Indicate relative importance by nesting elements and use instead for text with stress emphasis, <b>for text that is "stylistically offset" or bold without being more important, or <mark> for highlighting relevant text.

Code :

```
<!DOCTYPE html>
<html>
<body>

        <p>Get <strong>asthma inhaler!</strong>,<br>he has got asthma attack!</p>
</body>
</html>
```

Output :

The <small> Element

If you have forgotten what small does in html 4 then let me('me' as in this document!) remind you that "The HTML Small Element (<small>) makes the text font size one size smaller (for example, from large to medium, or from small to x-small) down to the browser's minimum font size."HTML5, this element is re purposed to represent side-comments and small print, including copyright and legal text, independent of its styled presentation.

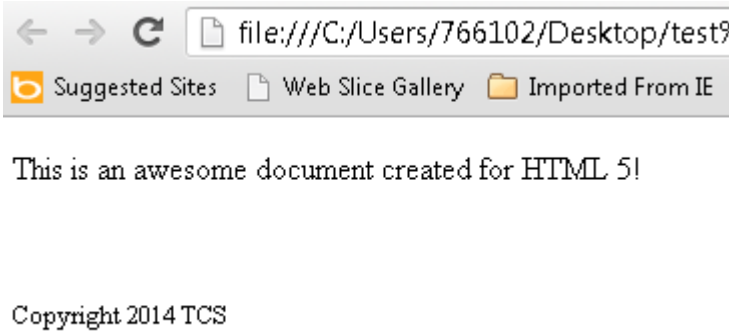
Don't use it for block-level content (paragraphs, lists, etc.), as this would be considered main content.

<small> text does not need to be smaller than surrounding text. If you just want smaller text, use CSS instead. Use <small> only on inline content. Finally, <small> doesn't affect the semantics of or .

Code :

```
<!DOCTYPE html>
<html>
<body>

        <p>This is an awesome document created for HTML 5!</p><br>
        <p><small>Copyright 2014 TCS</small></p>
</body>
</html>
```

Output :



1.4.3 New Semantic Elements of HTML 5

- **Time**

Tag Name : <time>

Description :
This is a newly introduced tag in HTML5.
It is used to specify human readable time and date format in an HTML document.
You can put anything you want between opening and closing tags. But machine-readable code is written inside attribute datetime.

Further explanation of the date is given in "**Dates**" tag in this document.

Attributes:

Attribute	Value	Description
datetime	datetime	Represents a machine readable date/time of time

		element.

Code ::

```
<time datetime="1905"> year 1905 </time>
```

Output::

year 1905

Code ::

```
<time datetime="1905-11-13"> 13th november 1905 </time>
```

Output:

13th november 1905

Another Example:

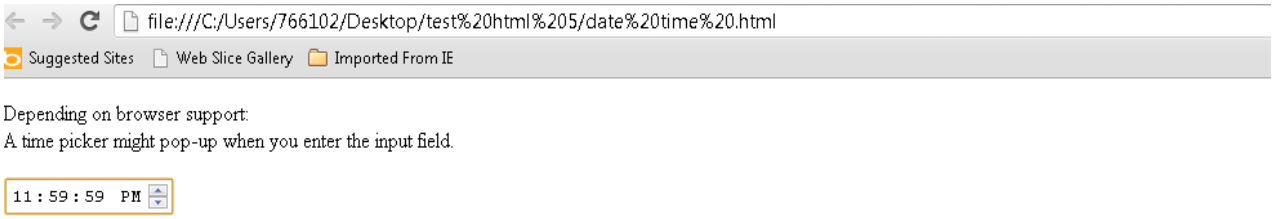
You can take time in from the user and then use that time to send it in backend to your application in the format you want using javascript. Let's see example for this scenario.

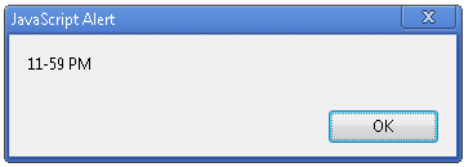
Code ::

```
<!DOCTYPE html>
<html>
<head>
<script type="text/javascript">
function format(){
var timeformat =document.getElementById("userTime").value;

var hour=timeformat.substring(0,2);
var min=timeformat.substring(3,5);
var time;
if(hour==12)
{
time=hour+"-"+min+" "+"PM";
}
else if(hour>12){
time=hour-12+"-"+min+" "+"PM";
}
else{
time=hour+"-"+min+" "+"AM";
}
```

```
alert(time);
}
</script>
</head>
<body>

<p>
Depending on browser support:<br>
A time picker might pop-up when you enter the input field.
</p>

<form>

  <input type="time" id="userTime" min="00:00:00" onblur="format()"/>

</form>
</body>
</html>
```

Output:





Real time Application : To set dates in human-readable format but pass machine-readable to computer

Browser Support :

Chrome	IE	Mozilla Firefox	Safari	Opera
6	9	4	5	11.1

- **Mark**

Tag Name: <mark>

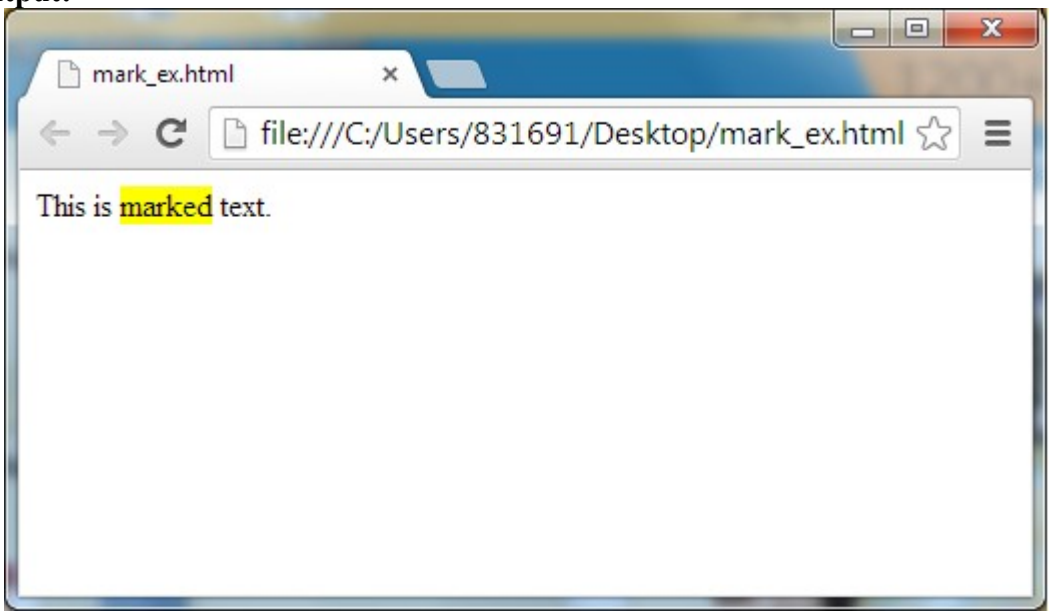
Description : The HTML <mark> tag is used for indicating text as marked or highlighted for reference purposes.

Attributes : Supports all global attributes and event attributes defined in conventional html.

Code: To highlight the word “marked” in whole line.

```
<p>This is <mark>marked</mark> text.</p>
```

Output:



Browser Support :

Element					
<mark>	6.0	9.0	4.0	5.0	11.1

1.4.4 Media Elements

- **Video**

Tag Name : <video>

Description : This is a newly introduced tag in HTML5. It can be used to specify video(i.e. Movie clip or stream) in an HTML document.

Attributes:

Attribute	Value	Description
src	URL	Specifies the URL of the video file.

autoplay	Autoplay	The mere presense of this attribute specifies that video should be played as soon as it is loaded.
loop	loop	Specifies whether video will start again or not.
controls	controls	Specifies that controls should be displayed or not.
preload	Auto Metadata none	Specifies how the video should be loaded when page loads.
height	pixels	Sets height of the video player
muted	Muted	Specifies that audio output of video should be muted or not
poster	URL	The url to the image you want to display till your video is loaded
width	pixels	Sets the width of your video player

Subtags: <source> : Defines multiple media resources for media elements.

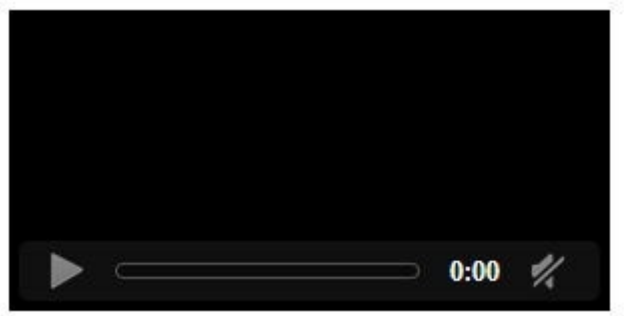
Events :

abort	Fires when loading is aborted
Canplay	Fires when video starts playing
canplay through	Fires when browser can play through video without buffering
Emptied	Fires when current playlist gets emptied.
Ended	Fires when playlist ends
Error	Fires when during loading of content any error comes.
Pause	Fire when video is paused.
play	Fires when video starts playing again
seeked	Fires when user is finished moving to the new position.
Stalled	Fires when browser tries to load data but data is unavaliable.
Timeupdate	Fires when playback position is changed
Volume change	Fires when the volume has changed

Code ::

```
<video autoplay="autoplay" controls>
      <source src="/media/video.ogg" type="video/ogg" />
     <source srt=”/media/video.mp4” type=”video/mp4”>

       Your browser does not support HTML5 video.
</video>
```

Snapshot::



Real time Application : To include any video file in your web applications.

Browser Support :
3 video formats are supported for <video> element currently MP4, WebM and Ogg

Browser	Mp4	WebM	Ogg
Internet Explorer	Yes	No	No
Chrome	Yes	Yes	Yes
Firefox	Yes	Yes	Yes
Safari	Yes	No	No
Opera	No	Yes	Yes

- **Audio**

Tag Name : <audio>

Description : The HTML <audio> tag is used to specify audio on an HTML document.

For example, you could embed a music file on your web page for your visitors to listen to.

Attributes: Attributes supported by Audio tag.

Attribute	Value	Description
src	URL	Specifies the location of the audio file.
autoplay	Boolean	Specifies whether or not to start playing the audio as soon as the object has loaded.
loop	Boolean	Specifies whether to keep re-playing the audio once it has finished.
controls	Boolean	Specifies whether or not to display audio controls (such as a play/pause button etc).
autobuffer	Boolean	the audio will automatically begin buffering even if it's not set to automatically play.

Events :

Event	Description
abort	This event is generated when playback is aborted.
canplay	This event is generated when enough data is available that the media can be played.
ended	This event is generated when playback completes.
error	This event is generated when an error occurs.
loadeddata	This event is generated when the first frame of the media has finished loading.
loadstart	This event is generated when loading of the media begins.
pause	This event is generated when playback is paused.
play	This event is generated when playback starts or resumes.
progress	This event is generated periodically to inform the progress of the downloading the media.
ratechange	This event is generated when the playback speed changes.
seeked	This event is generated when a seek operation completes.
seeking	This event is generated when a seek operation begins.
suspend	This event is generated when loading of the media is suspended.
volumechange	This event is generated when the audio volume changes.
waiting	This event is generated when the requested operation (such as playback) is delayed pending the completion of another operation (such as a seek).

Code :

```
<audio autoplay="autoplay" controls>
      <source src="/html5/audio.ogg" type="audio/ogg" />

       Your browser does not support the <audio> element.
</audio>
```

Output:

Real time Application : To include any audio file in your web applications.

Browser Support :

Browser	Mp3	Wav	Ogg
Internet Explorer	Yes	No	No
Chrome	Yes	Yes	Yes
Firefox	Yes	Yes	Yes
Safari	Yes	Yes	No
Opera	No	Yes	Yes

• Embed

Tag Name: <Embed>

Description : The HTML <embed> tag represents a container for external application or interactive content.

Attributes: Attributes supported by embed tag.

Attribute	Value	Description
Height	pixels	Specifies the height.
src	URL	Specifies the address of the source file.
type	MIME_type	Specifies the MIME type.
width	pixels	Specifies the width.

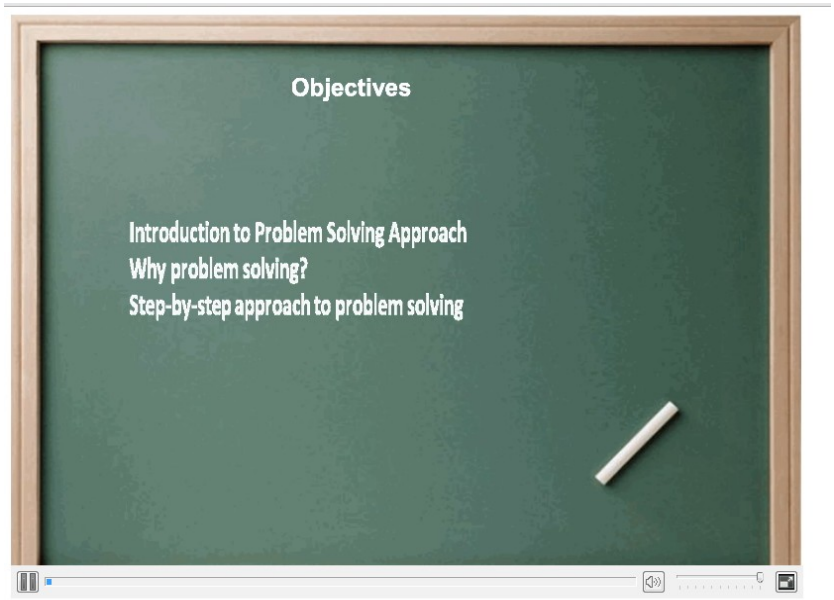
HTML5 Event Attributes		
onabort	onblur	oncanplay
oncanplaythrough	onchange	onclick
oncontextmenu	ondblclick	ondrag
ondragend	ondragenter	ondragleave
ondragover	ondragstart	ondrop
ondurationchange	onemptied	onended
onerror	onfocus	onformchange
onforminput	oninput	oninvalid
onkeydown	onkeypress	onkeyup
onload	onloadeddata	onloadedmetadata
onloadstart	onmousedown	onmousemove
onmouseout	onmouseover	onmouseup
onmousewheel	onpause	onplay
onplaying	onprogress	onratechange
onreadystatechange	onscroll	onseeked
onseeking	onselect	onshow
onstalled	onsubmit	onsuspend
ontimeupdate	onvolumechange	onwaiting

Code: To embed videlan VLC Media Player plugin in google chrome

```
<embed  type='application/x-vlc-plugin'pluginspage='http://www.videolan.org'
        version='VideoLAN.VLCPlugin.2'
        width='800'
```

```
            height='600'
            id='vlc'
            loop='yes'
            autoplay='yes'
            target='sourceLocation(can be remote or local)'>
        </embed>
```

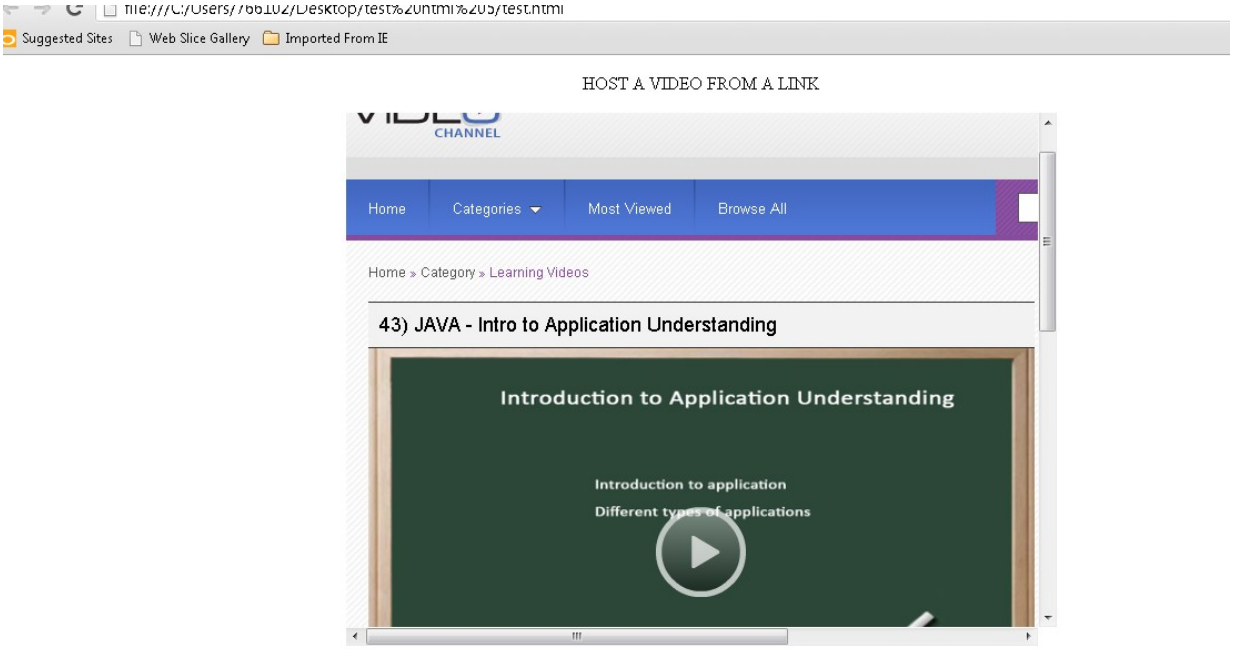
Output:



Next Example :

```
<!DOCTYPE html>
<html>
<body>
<div align="center">
<p>HOST A VIDEO FROM A LINK</p>
<embed src="http://video.tcs.com/play?
nid=4301javaintrotoappunderstanding&context=_nfpb&_pageLabel=l_logo"
height="480" width="640"  type="video/mp4"/>
</div>
</body>
</html>
```

Output:



Real time Application : To include any third party applications of any plugins into your application. For example, VLC Media player plugin or Flash game plugin

Browser Support :

Element					
<embed>	Yes	Yes	Yes	Yes	Yes

1.4.5 HTML 5 Form and Form Elements

No doubt that you always interact with at least one form if you it's web based application. HTML5 introduces a number of new attributes, input types, and other elements to our markup toolkit. We will see new features provided by HTML 5 now.

1.4.5.1 HTML 5 Form attributes

Now we will cover few of the new attributes that we can add to our forms and start using today. Not only will they improve your users' experience, but they will save you from having to write a lot of JavaScript.

- **placeholder**

Description:

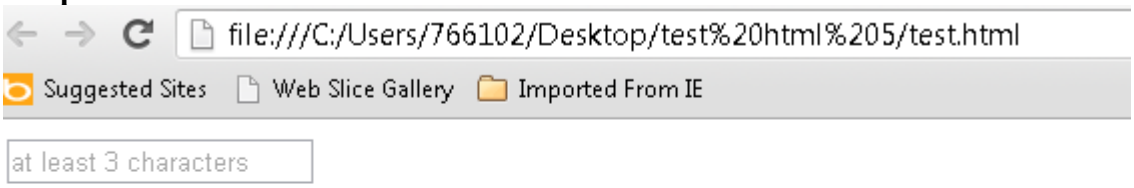
First is the placeholder attribute, which allows us to set placeholder text as we would currently do in HTML4 with the value attribute. It should only be used for short descriptions. For anything longer, use the title attribute. The difference from HTML4 is that the text is only displayed when the field is empty and hasn't received focus. Once the field receives focus (e.g., you click or tab to the field), and you begin to type, the text simply disappears.

Browsers that don't support the placeholder attribute ignore it, so it won't render. By including it, though, you're providing an enhanced user experience for those users who have browser support and you're "future proofing" your site. All modern browsers support placeholder and it will also be supported in Internet Explorer 10.

Code:

```
<input type="text" name="user-name" id="user-name" placeholder="at least 3 characters">
```

Output:



- **autofocus**

Description:
Autofocus does exactly what it says. Adding it to an input automatically focuses that field when the page is rendered. As with placeholder, autofocus is something that we used JavaScript for in the past.

Traditional JavaScript methods do, though, have some serious usability problems. For example, if a user begins completing the form before the script loads, they will be returned to the first form field when the script is loaded. The autofocus attribute in HTML5 gets around this issue by focusing as soon as the document loads, without having to wait for the JavaScript to be loaded.

It is a Boolean attribute (except if you are writing XHTML5,you can explore what I mean to say(again I mean "this doc") and is implemented as follows:

All modern browsers support the attribute and, like placeholder, browsers that don't support the autofocus attribute simply ignore it.

Note Several new HTML5 form attributes are Boolean attributes. This just means they're set if they're present and not set if they're absent. They can be written several ways in HTML5.

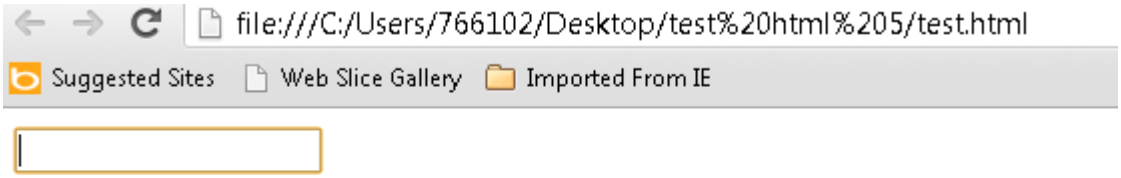
autofocus
autofocus=""
autofocus="autofocus"

However, if you are writing XHTML5, you have to use the autofocus="autofocus" style

Code:

```
<input type="text" name="first-name" id="first-name" autofocus>
```

Output:



- **autocomplete**

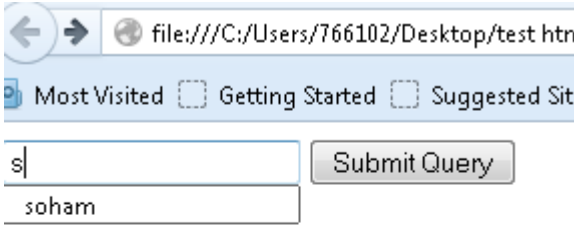
Description:
The autocomplete attribute helps users complete forms based on earlier input. The attribute has been around since IE5.5 but has finally been standardized as part of HTML5. The default state is set to on. This means that generally we won't have to use it. **However, if you want to insist that a form field be entered each time a form is completed (as opposed to the browser autofilling the field), you would implement it like so:**

Code:

```
<input type="text" name="tracking-code" id="tracking-code" autocomplete="off">
```

Below given example remove autocomplete=”off”,also sometimes you need to do browser setting as well.

Output:



- **required**

Description:
The required attribute doesn't need much introduction; like autofocus, it does exactly what you'd expect. By adding it to a form field, the browser requires the user to enter data into that field before submitting the form. This replaces the basic form validation currently implemented with JavaScript, making things a little more usable and saving us a little more development time. required is a Boolean attribute, like autofocus. Let's see it in action.

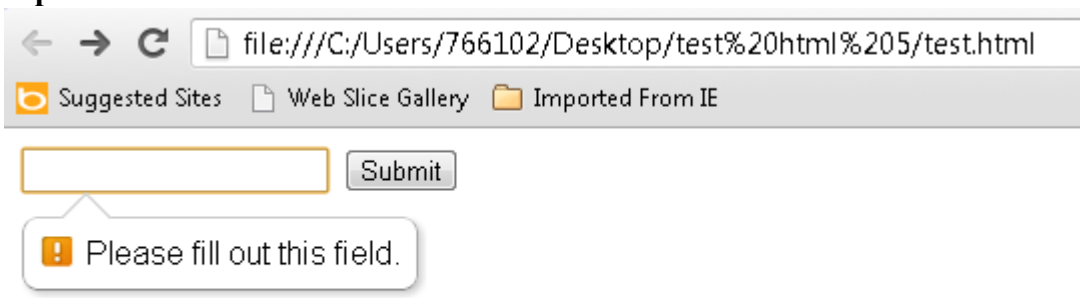
required is currently implemented only in Opera 9.5+, Firefox 4+, Safari 5+, and Chrome 5+, so for the time being you need to continue writing a script to check that fields are completed on the client side in other browsers (IE!). Internet Explorer 10 will support required. Opera, Chrome, and Firefox show the user an error message upon form submission. In most browsers, the errors are then localized based on the declared language. Safari doesn't show an error message on submit, but instead places focus on that field.

The default rendering of "required" error messages depends on the individual browser; at present the error message bubble can't be styled with CSS in all browsers. Chrome, however, does have a proprietary property that you can use to style the error bubble. You can find it out how to do that!

Code:

```
<input type="text" id="fist-name" name="first-name" required>
```

Output:



- **pattern**

Description:
The pattern attribute is likely to get a lot of developers very excited. It specifies a JavaScript regular expression for the field's value to be checked against. pattern makes it easy for us to implement specific validation for numeric validation which is some specific numbers etc, and so on. The possibilities for pattern are wide-ranging, and this is just one simple example using a product number.

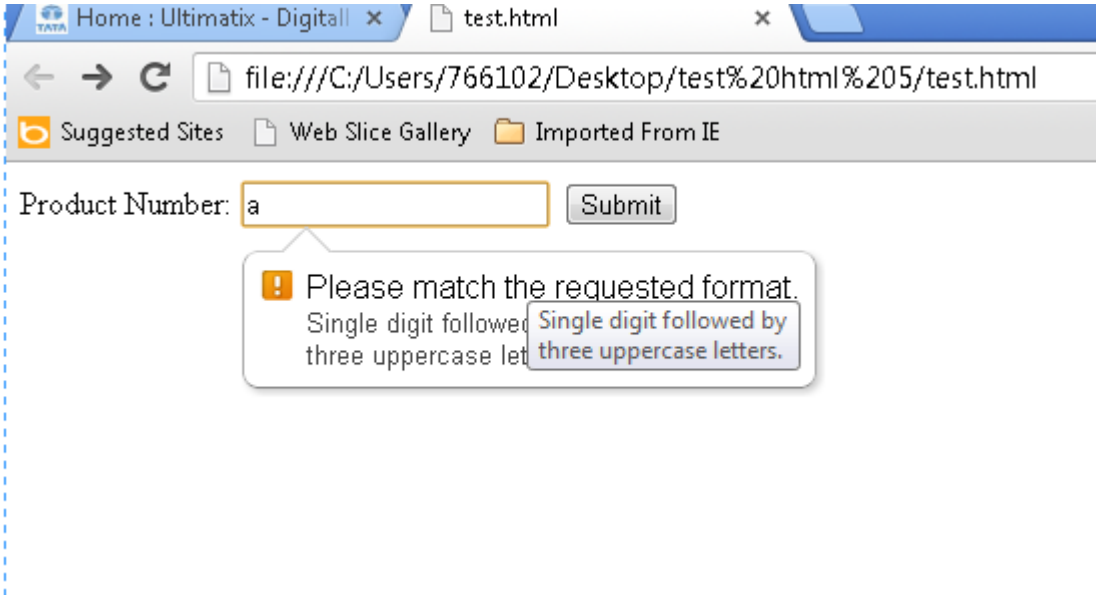
In the given example the pattern describes that the product number should be a single digit [0-9] followed by three uppercase letters [A-Z]{3}.You can always create customized patterns as well.

As with required, Opera 9.5+, Firefox 4+, Safari 5+, and Chrome 5+ are the only browsers with support for pattern at present. Internet Explorer 10 will support pattern. However, with the browser market moving at a fast pace, the others will soon catch up.

Code:

```
<label> Product Number:
<input pattern="[0-9][A-Z]{3}" name="product" type="text" title="Single digit followed by
three uppercase letters."/>
</label>
```

Output:



- **List and the Datalist Element**

Description:
The list attribute enables the user to associate a list of options with a particular field. The value of the list attribute must be the same as the ID of a datalist element that resides in the same document. The datalist element is new in HTML5 and represents a predefined list of options for form controls. It works in a similar way to the in-browser search boxes that autocomplete as you have just seen autocomplete.

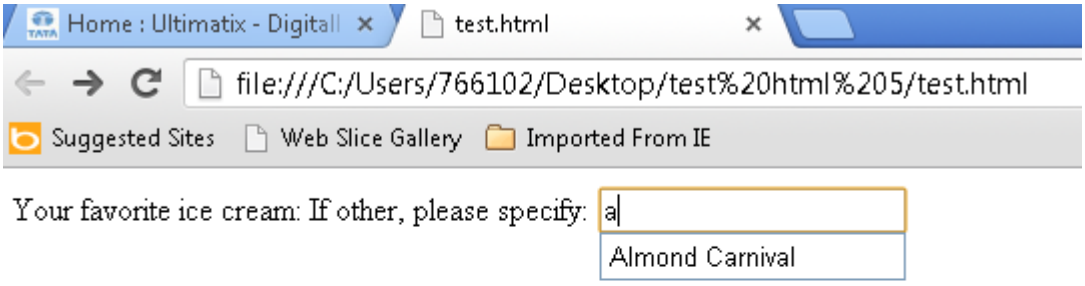
Browser support for list and datalist is currently limited to Opera 9.5+ and Firefox 5+. Internet Explorer 10 will support the list attribute and datalist.

Code:

```
<!DOCTYPE html>
<form name="myForm" action="/" method="get">

<label>Your favorite ice cream:
<datalist id="icecream">
   <option value="va">Vannila</option>
   <option value="st">Strawberry</option>
   <option value="ac">Almond Carnival</option>
   <!-- ... -->
 </datalist>
If other, please specify:
```

```
<input type="text" name="icecream" list="icecream">
</label>

</form>
```

Output:



- **form(Remember we are talking about attributes and not tag)**

Description:
The form attribute is used to associate an input, select, or textarea element with a form (known as its form owner). Using form means that the element doesn't need to be a child of the associated form and can be moved away from it in the source. The primary use case for this is that input buttons that are placed within tables can now be associated with a form.

Code:

```
<!DOCTYPE html>

<input type="button" name="sort-l-h" form="sort">
```

Output: Output can be interpreted by yourself.

- ## formaction

Description:
formaction specifies the file or application that will submit the form. It has the same effect as the action attribute on the form element and can only be used with a submit or image button (type="submit" or type="image"). When the form is submitted, the browser first checks for a formaction attribute; if that isn't present, it proceeds to look for an action attribute on the form.

Code:
```
<!DOCTYPE html>

<input type="submit" value="Submit" formaction="abc.html">
```

Output: Output can be interpreted by yourself.

- ## formenctype

Description:
formenctype details how the form data is encoded with the POST method type. It has the same effect as the enctype attribute on the form element and can only be used with a submit or image button (type="submit" or type="image"). The default value if not included is application/x-www-form-urlencoded.

Code:
```
<!DOCTYPE html>

<input type="submit" value="Submit" formenctype="application/x-www-form-urlencoded">
```

Output: Output can be interpreted by yourself.

- ## formmethod

Description:
formmethod specifies which HTTP method (GET, POST, PUT, DELETE) will be used to submit the form data. It has the same effect as the method attribute on the form element and can only be used with a submit or image button (type="submit" or type="image").

Code:

```
<!DOCTYPE html>

<input type="submit" value="Submit" formmethod="POST">
```

Output: Output can be interpreted by yourself.

- **formtarget**

Description:
formtarget specifies the target window for the form results. It has the same effect as the target attribute on the form element and can only be used with a submit or image button (type="submit" or type="image").

Code:

```
<!DOCTYPE html>

<input type="submit" value="Submit" formtarget="_self">
```

Output: Output can be interpreted by yourself.

1.4.5.2 HTML 5 Form New Input Types

Now we will look few new input types(around dozen) which are been introduced in html 5.

- **Search**

Description:
When we talk about search, we're not just talking about Google search. We are talking about the search field on that ecommerce site you just made a purchase from. It's probably the most common action performed on the Web every day, yet it's not marked up very semantically, is it? We all tend to write something like this:

<input type="text" name="search">

Well, we can write in html 5 is

Code:

```
<!DOCTYPE html>
```

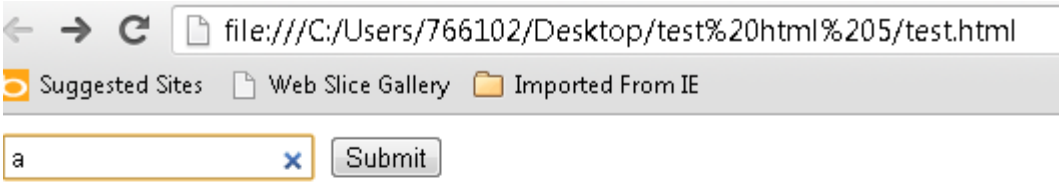
```
<input type="search" name="search">
```

Desktop browsers will render this in a similar way to a standard text field—until you start typing, that is. At this point, a small cross appears on the right side of the field. Notice the x in output. This lets us quickly clear the field!

Now if it is mobile device then when you click on that search field, you can notice the keyboard in which at bottom right instead of normal "go" text it would display "Search",Now you will say what is a big deal in that! Well, to display appropriate things and to ensure the smoother experience is the main aim of html 5!!Think that your effort to code the same thing has been saved!!

If a browser doesn't understand type="search", it will default to type="text". This means you're not losing anything. This applicable to all the new input types.

Output:



- **email**

Description:
The email input type is no different from a standard text input type and allows for one or more e-mail addresses to be entered. Combined with the required attribute, the browser is then able to look for patterns to ensure a valid e-mail address has been entered. Opera 9.5+, Firefox 4+, and Chrome 5+ have already implemented this basic validation, with support coming in IE 10. The browser goes as far as presenting the user with an error message if the email address

entered isn't valid.

Code:

```
<!DOCTYPE html>
<input type="email" name="email" required>
```

Output:

- **URL**

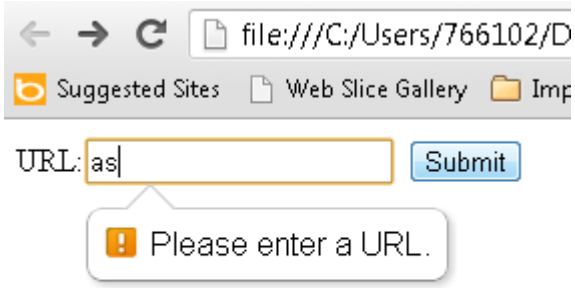
Description:
The url input type, as you might expect, is for web addresses. You can use the multiple attribute to enter more than one URL. Like type="email", a browser will carry out simple validation on these fields and present an error message on form submission. This is likely to include looking for forward slashes, periods, and spaces, and possibly detecting a valid top-level domain (such as .com or .co.in).

There is currently some debate about whether the user has to enter http:// into a field using input="url" . Browser vendors are discussing the possibility of pre-pending http:// to a URL following a check upon form submission. For up-to-date information,you can always refer to the HTML5 specification.

Code:

```
<!DOCTYPE html>
<input type="url" name="url" required>
```

Output:



- **tel**

Description:
tel differs from email and url in that no particular syntax is enforced. Phone numbers differ around the world, making it difficult to guarantee any type of specific notation except for allowing only numbers and perhaps a + symbol to be entered. It's possible that you can validate specific phone numbers (if you can guarantee the format) using client-side validation.

Code:

```
<!DOCTYPE html>
<input type="tel" name="tel" id="tel" required>
```

Output: Output can be interpreted by yourself.

- **number**

Description:
number, as you might expect, is used for specifying a numerical value. As with the majority of these new input types, Opera was the first to implement type="number". It, Safari, and Chrome render the input as a spinbox control whereby you can click the arrows to move up or down. Or if you prefer, you can type directly into the field. Firefox, on the other hand, renders the field like a standard text box. Support for type="number" will be in IE 10 also.

With the additional attributes min, max, and step, we can change the default step value of this spinbox control as well as set minimum, maximum, and starting values (using the standard HTML value attribute). Given example shows how these attributes work:

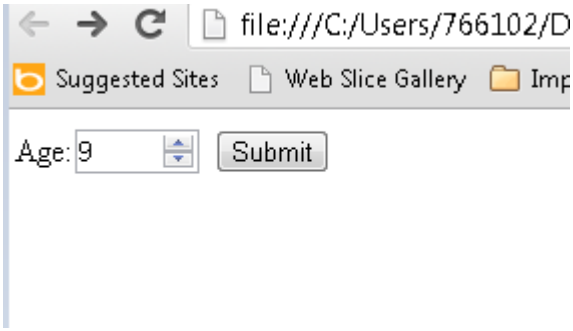
In this example, min represents the minimum value the field will accept, and max represents the maximum value. If we reach the maximum or minimum value, the appropriate arrow on the spinbox control will be greyed out so you can no longer interact with it. step is the increment that the value should adjust up or down, with the default step value being 1. This means we can include negative values or step up in increments of 0.5 or 5. value is that attribute you're used to from previous versions of HTML. Each of the attributes are optional, with defaults being set if they aren't used.

In contrast to Opera's implementation, the iPhone and some Android devices simply render the field as a standard text box but optimize the keyboard(will show only numbers) for easy input.

Code:

```
<!DOCTYPE html>

Age:<input type="number" min="5" max="18" step="1" value="9" name="age" >
```

Output:



- **Dates**

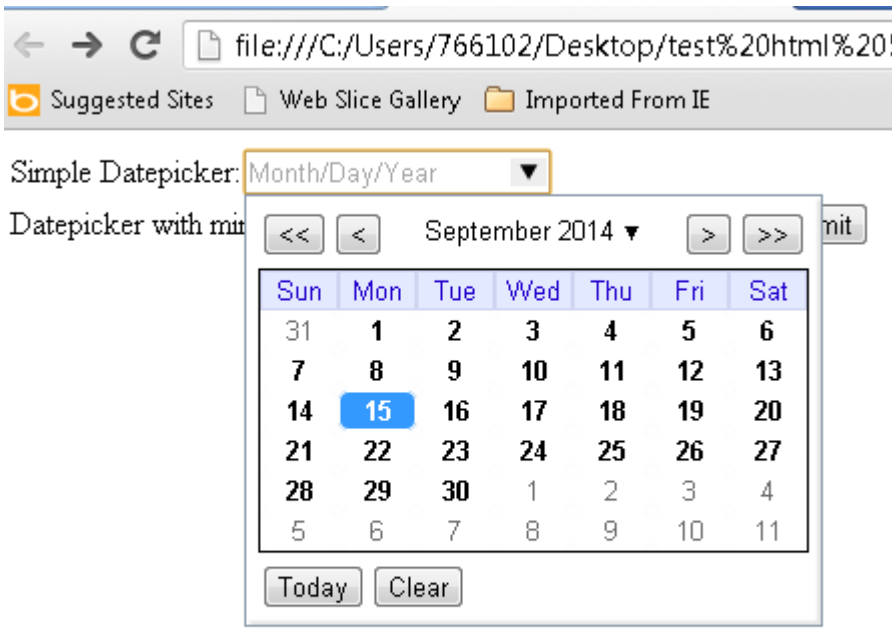
Description:
If you've ever booked tickets online, you will have come across date pickers to help you quickly and easily choose the date you require. Perhaps you've even implemented a date picker in your web application. Generally, this is done using a JavaScript library such as jQuery, Dojo, or YUI. It can be a pain when you need to load a whole library and associated plug-ins just to implement a simple date picker. Well, with HTML5 we get that functionality baked into the browser. Not only that, but we don't have to stop at just selecting a single date; we can select a week, month, time, date and time, and even date and time with a time zone using the different input types. You can explore those as per your requirements. The markup is pretty straightforward.

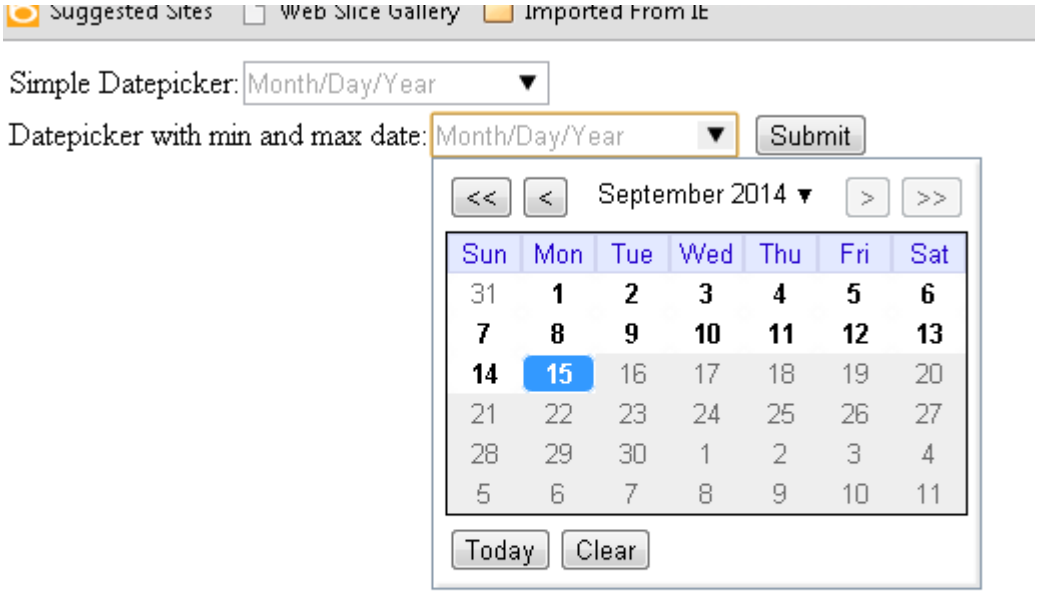
You can go a step further by using the min and max attributes to ensure the user can only choose from a specified date range.

Code:

```
<!DOCTYPE html>

<input id="dob" name="dob" type="date">

<input id="startdate" name="startdate" min="2012-01-01" max="2013-01-01"
type="date">
```

Output:


file:///C:/Users/766102/Desktop/test%20html%20
Suggested Sites
Web Slice Gallery
Imported From IE
Simple Datepicker:
Month/Day/Year
Datepicker with mi
mit
<<
<
September 2014
>
>>
Sun Mon Tue Wed Thu Fri Sat
31 1 2 3 4 5 6
7 8 9 10 11 12 13
14 15 16 17 18 19 20
21 22 23 24 25 26 27
28 29 30 1 2 3 4
5 6 7 8 9 10 11
Today
Clear



Another Example:

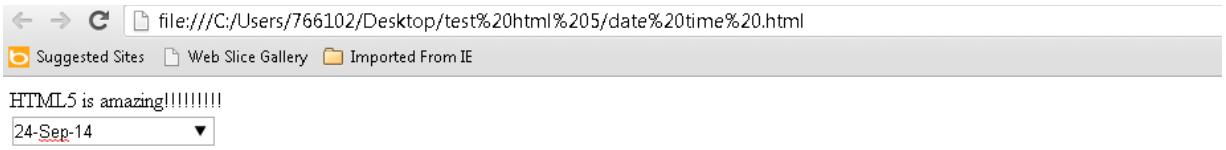
You can manipulate date format based on your requirement using javascript and use required format in your backend. Let's see example for this scenario.

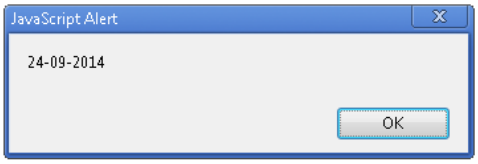
Code:

```
<!DOCTYPE html>
<html>
<head>
<script type="text/javascript">
function format(){
        var fulldate= document.getElementById("fulldate").value;
        var year=fulldate.substring(0,4);
        var month=fulldate.substring(5,7);
        var date=fulldate.substring(8);
        var date=date+"-"+month+"-"+year;
        alert(date);
}
</script></head>
<body>

       <!--Date format implementation using HTML 5  -->

   HTML5 is amazing!!!!!!!!!
   <form>
    <input type="date" id="fulldate"  onblur="format()"  />
```

```
</form>
```

Output:


file:///C:/Users/766102/Desktop/test%20html%205/date%20time%20.html
Suggested Sites
Web Slice Gallery
Imported From IE
HTML5 is amazing!!!!!!!!!
24-Sep-14


JavaScript Alert
24-09-2014
OK

1.4.5.3 HTML 5 Interesting Form Elements Elements For Web Application

- **output(This time it is the name of the element)**

Description:
The output element is simple; it represents the result of a calculation or user action. Using the for attribute, a relationship is created between the result of the calculation and the value(s) that went into the calculation.

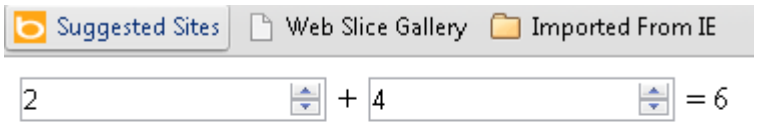
The output element is supported in Opera 9.5+, Firefox 4+, Chrome, Safari 5+, and Internet Explorer 10. The given example adds the values of two inputs together to create the output.

This is triggered by the oninput event placed on the output element and handled by the browser using the new valueAsNumber function—no complex scripting required!

Code:

```
<!DOCTYPE html>
<form onsubmit="return false" oninput="result.value = firstNumber.valueAsNumber +
secondNumber.valueAsNumber">
  <input name="firstNumber" id="firstNumber" type="number" step="any"> +
  <input name="secondNumber" id="secondNumber" type="number" step="any"> =
  <output name="result" for="firstNumber secondNumber"></output>

</form>
```

Output:



- **progress**

Description:
The <progress> tag represents the progress of a task,it can be seen as a more specific kind of meter element that's only used to measure progress of a task While the specifics of how it's displayed is left up to the developer, it's typically displayed as a progress bar. Javascript can be used to manipulate the value of progress bar.

Attributes:
In addition to the global attributes, this tag also supports the following attributes:

1.max:
This attribute describes how much work the task indicated by the progress element requires. The max attribute, if present, must have a value greater than zero and be a valid floating point number.

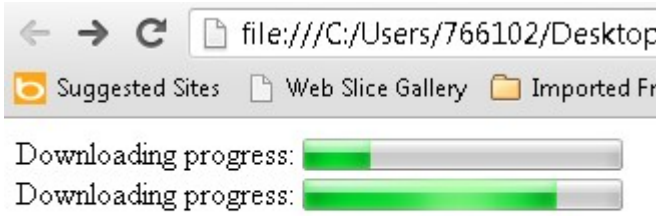
2.value:
This attribute specifies how much of the task that has been completed. It must be a valid

floating point number between 0 and max, or between 0 and 1 if max is omitted. If there is no value attribute, the progress bar is indeterminate; this indicates that an activity is ongoing with no indication of how long it is expected to take.

Simple Code:

```
<!DOCTYPE html>
<html>
<body>

   Downloading progress:
  <progress value="22" max="100">
  </progress><br>

   Downloading progress:
  <progress value="0.8">
   </progress>

</body>
</html>

</form>
```

Output:

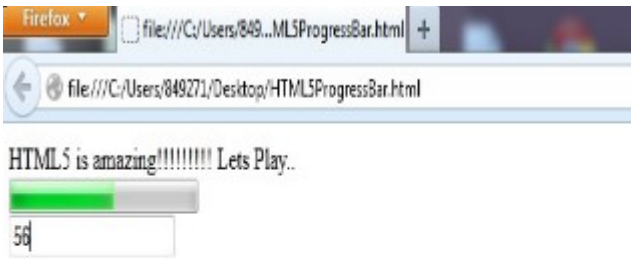


Dynamic Code For Progress:

```
<!DOCTYPE html>
<html>
```

```
<head>

<script type="text/javascript">
function progress(){
        var progress= document.getElementById("input1").value;
        document.getElementById("progress").value=progress;
}
</script></head>
<body>
      <!--Progress Bar simple fun using HTML 5  -->

   HTML5 is amazing!!!!!!!!! Lets Play..
   <form>

  <progress value="0" max="100"  id="progress" >

  </progress><br>

    <input type="text" id="input1" onkeyup="progress()"   />
  </form>
 </body>
</html>
```

Output:



- **meter**

Description:
Not to be confused with progress the matter element represents a scalar measurement within a known range, or a fractional value. Examples of this are the percentage completion of a social networking profile or the relevance of a particular search result. The meter element should not be used for arbitrary values such as height or weight unless there is a maximum known value. Using the high, low, max, and min attributes, the color of the element will change depending on the current value. Typically the value will be written server side prior to page load, but it may also be the case that the value is updated using JavaScript. This is also known as a gauge.

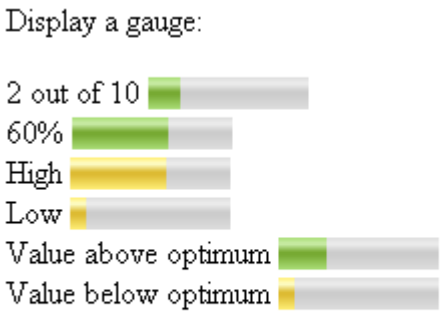
The meter element is supported in Opera 11+ and Chrome 7+.

Code:

```
<!DOCTYPE html>
<html>
<body>

        <p>Display a gauge:</p>
        2 out of 10 <meter value="2" min="0" max="10">2 out of 10</meter><br>

        60% <meter value="0.6">60%</meter><br>

        High   <meter value="6" min="0" max="10" high="5" low="3"></meter><br>

        Low   <meter value="1" min="0" max="10" high="5" low="3"></meter><br>

        Value above optimum   <meter value="0.3" high="0.9" low="0.2"
```

```
        optimum="0.5"></meter><br>

        Value below optimum  <meter value="0.1" high="0.9" low="0.2"
        optimum="0.5"></meter><br>

</body>
</html>
```

Output:



Dynamic Code For Progress:

```
<!DOCTYPE html>
<html>
<head>
<script type="text/javascript">
function meter(){
        var meter= document.getElementById("meterinput").value;
        document.getElementById("meter").value=meter;
}
</script>
</head>
<body>

        <p>Display a meter gauge:</p>

        Meter   <meter value="1" min="0" max="100" high="60" low="20" id="meter"
></meter><input type="text" id="meterinput" onkeyup="meter()" /><br>
</body>
</html>
```

Output:

- **details and summary**

Description:

The <details> tag specifies additional details that the user can view or hide on demand.
The <details> tag can be used to create an interactive widget that the user can open and close.
Any sort of content can be put inside the <details> tag.
The content of a <details> element should not be visible unless the open attribute is set.

Because details is generally paired with summary, let see what is summary!

The summary element represents a summary, caption, or legend for the contents of the summary element's parent details element, if any. summary must always be the first child of details, but more than one summary element can be used per details element.

Code:
Without open attribute

```
<!DOCTYPE html>
<html>
<head>
<meta http-equiv="Content-Type" content="text/html; charset=ISO-8859-1">
<title>HTML5 article example</title>
</head>
<body>
        <details>
  <summary>Copyright 1999-2014.</summary>
  <p> - by Refsnes Data. All Rights Reserved.</p>
  <p>All content and graphics on this web site are the property of the company Refsnes
Data.</p>
</details>
</body>
</html>
```

Output:

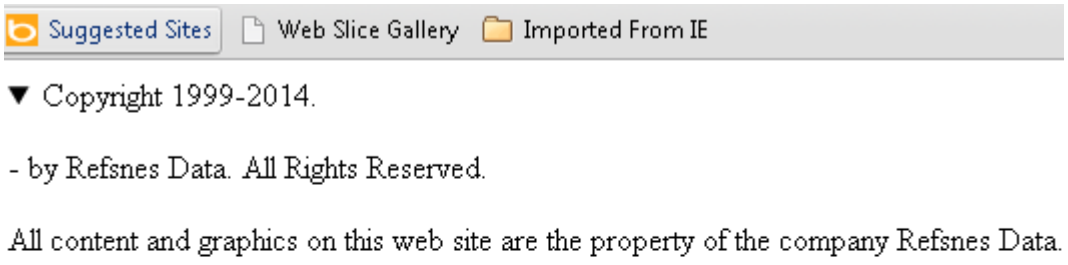
▶ Copyright 1999-2014.

Code:
With open attribute

```
<!DOCTYPE html>
<html>
<head>
<meta http-equiv="Content-Type" content="text/html; charset=ISO-8859-1">
<title>HTML5 article example</title>
</head>
<body>
```

```
        <details open="open">
 <summary>Copyright 1999-2014.</summary>
 <p> - by Refsnes Data. All Rights Reserved.</p>
 <p>All content and graphics on this web site are the property of the company Refsnes
Data.</p>
</details>
</body>
</html>
```

Output:



1.5 Summary(Now this time it is this document's summary!)

In the document we have focused on the importance of building a solid, structural foundation in HTML, creating a semantic layer upon which to build. We've introduced you to HTML5, the next generation of markup for the Web, and we've highlighted the benefits HTML5 has to offer and a large number of new, semantic elements that will ensure your web pages are just that extra bit future-friendly. So start implementing all these tags now!!